UNITED STATES INTERNATIONAL UNIVERSITY

AFRICA

DSA 3050A: Mid-Semester Practical Examination

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Spring Semester 2026

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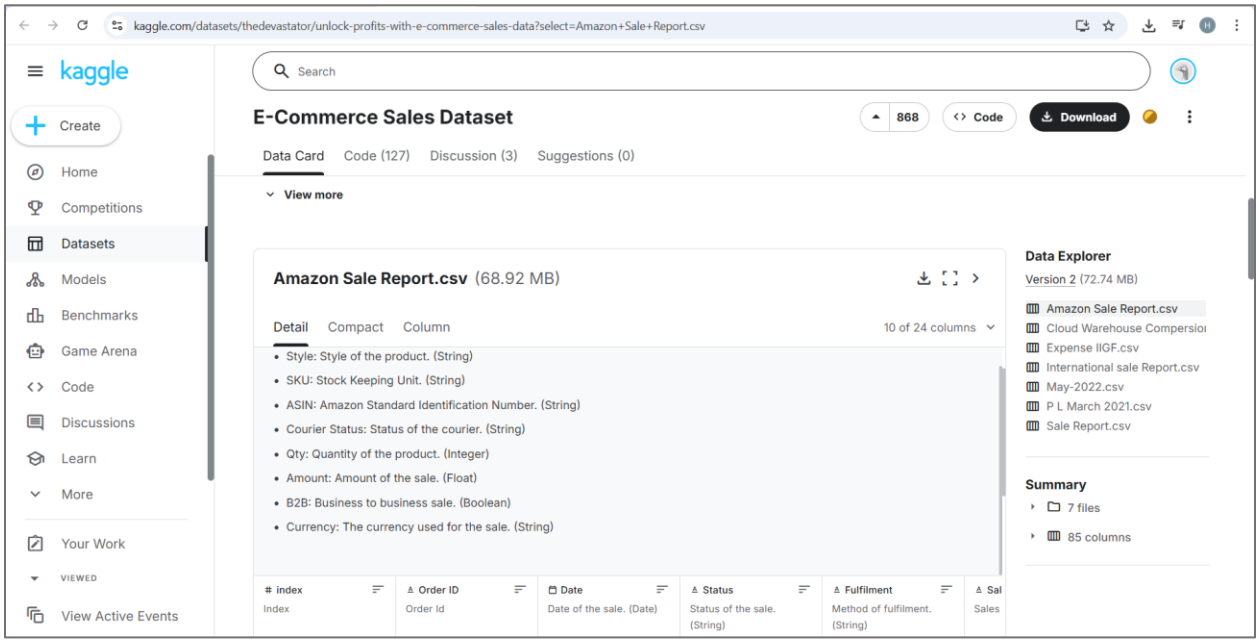
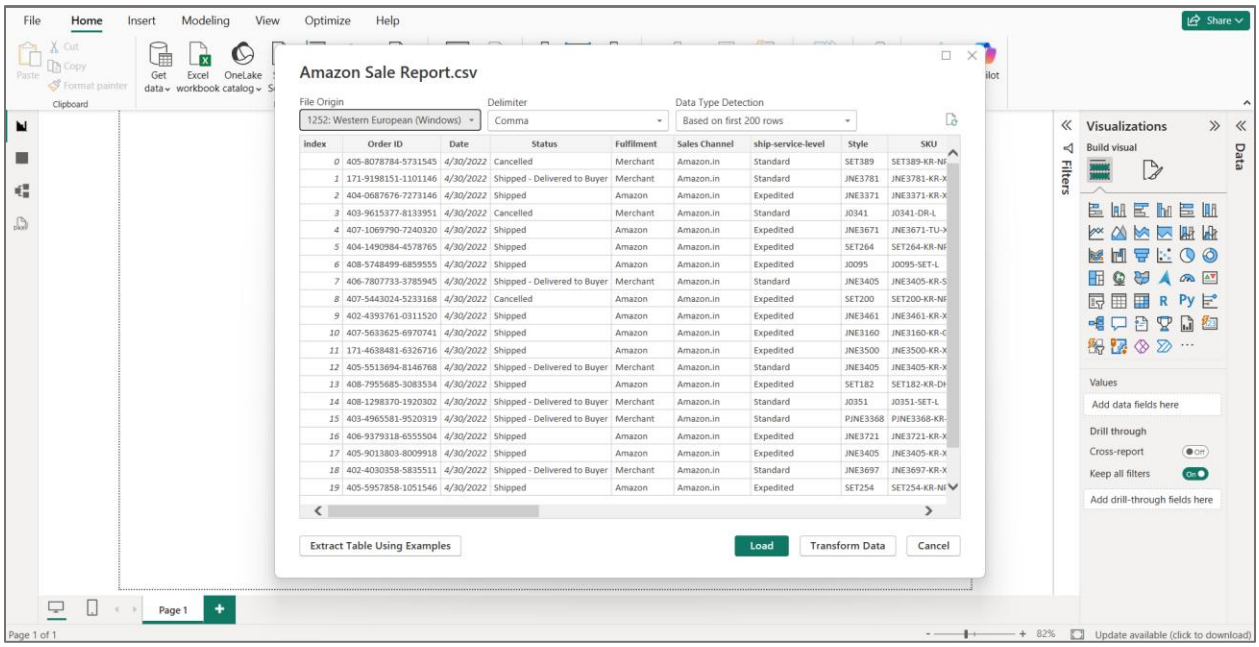
Dataset Source

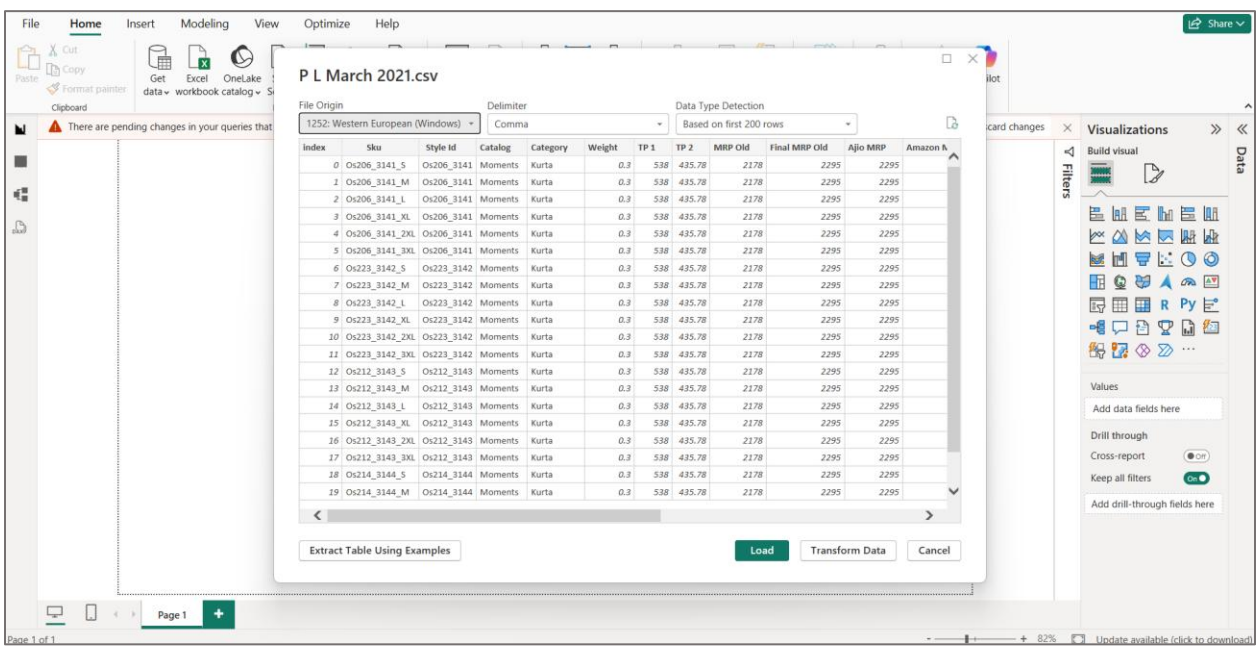
Real-world E-commerce Sales Dataset (7,000+ rows)

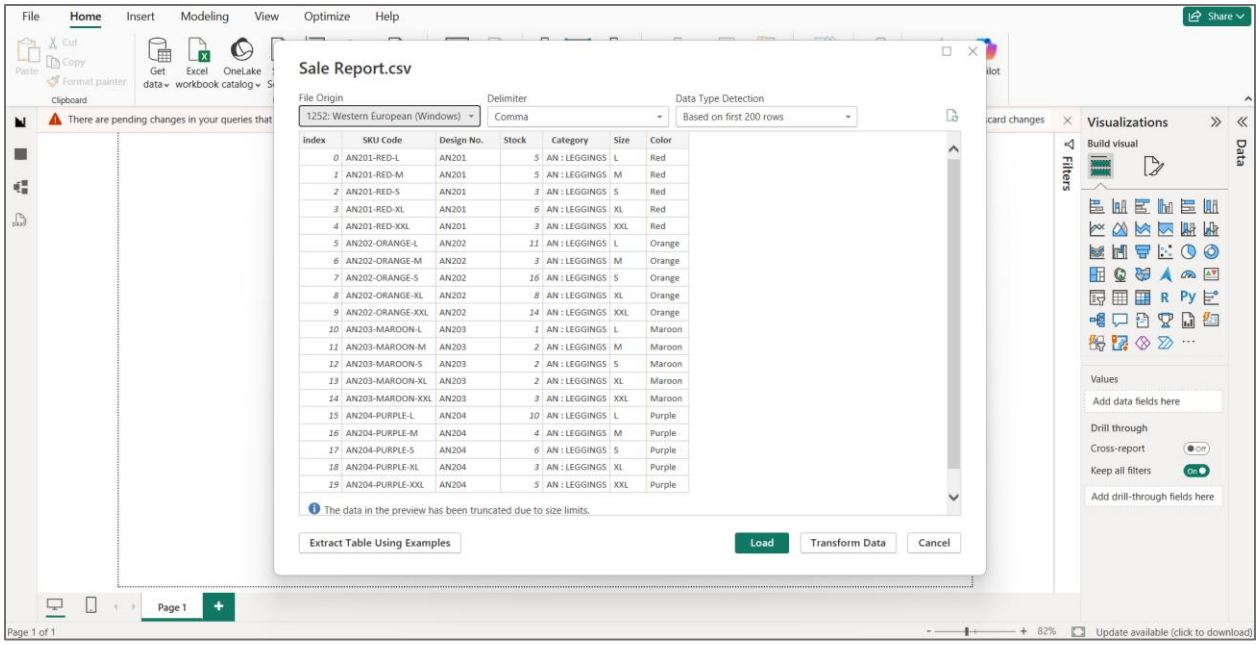


Section 1 - Process Evidence

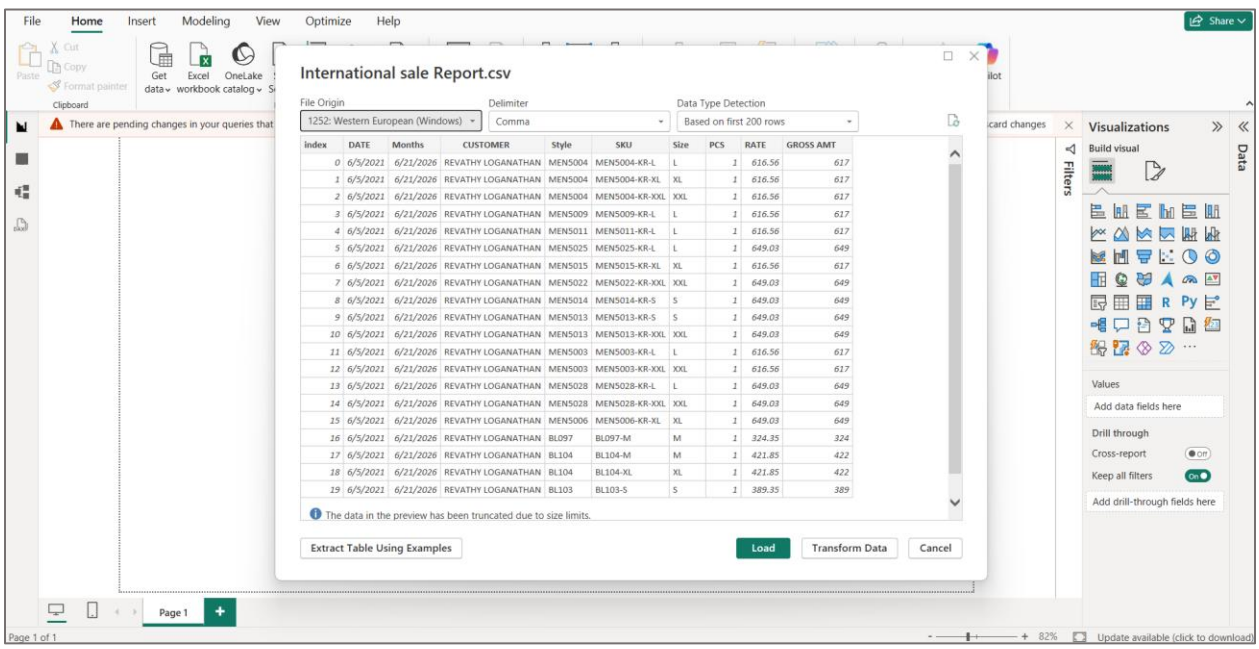
Dataset Loading





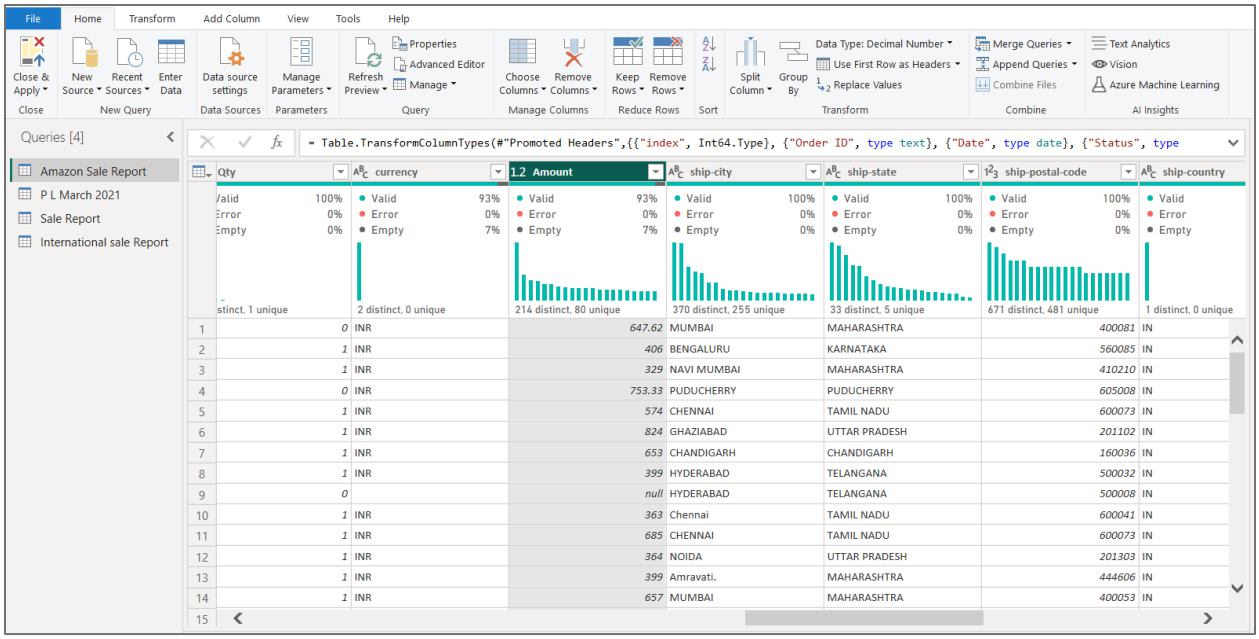


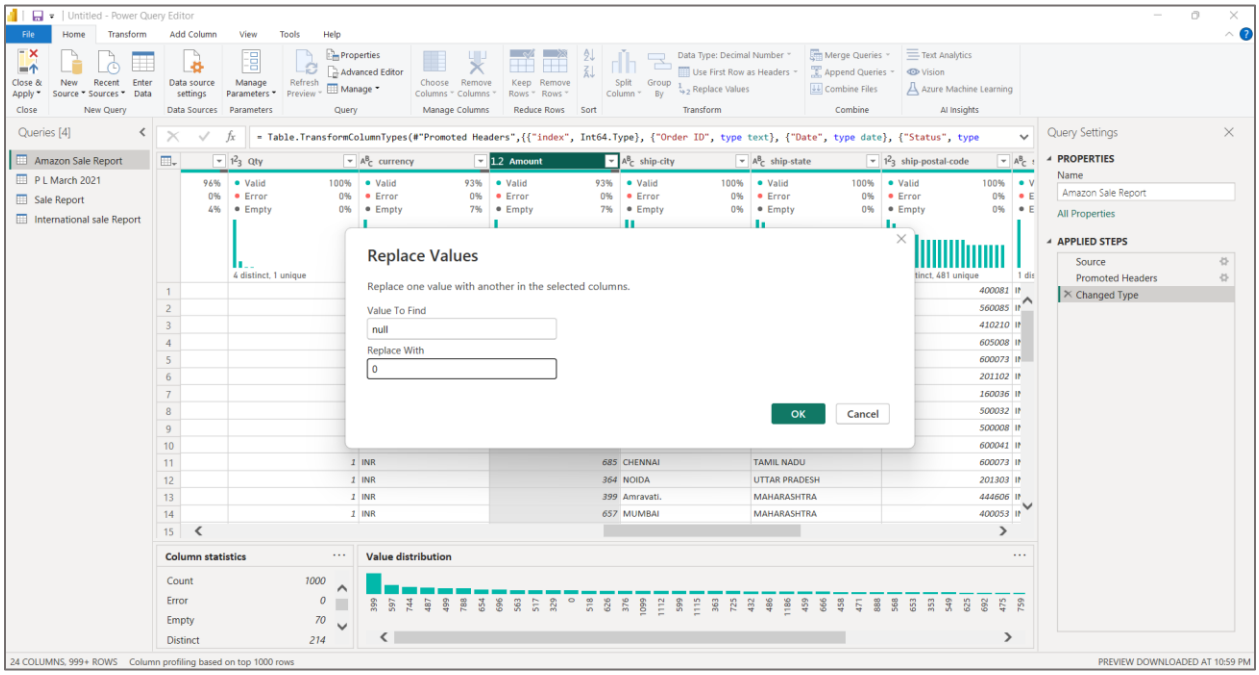
index	SKU Code	Design No.	Stock	Category	Size	Color
0	AN201-RED-L	AN201	5	AN : LEGGINGS	L	Red
1	AN201-RED-M	AN201	5	AN : LEGGINGS	M	Red
2	AN201-RED-S	AN201	3	AN : LEGGINGS	S	Red
3	AN201-RED-XL	AN201	6	AN : LEGGINGS	XL	Red
4	AN201-RED-XXL	AN201	3	AN : LEGGINGS	XXL	Red
5	AN202-ORANGE-L	AN202	11	AN : LEGGINGS	L	Orange
6	AN202-ORANGE-M	AN202	3	AN : LEGGINGS	M	Orange
7	AN202-ORANGE-S	AN202	16	AN : LEGGINGS	S	Orange
8	AN202-ORANGE-XL	AN202	8	AN : LEGGINGS	XL	Orange
9	AN202-ORANGE-XXL	AN202	14	AN : LEGGINGS	XXL	Orange
10	AN203-MAROON-L	AN203	1	AN : LEGGINGS	L	Maroon
11	AN203-MAROON-M	AN203	2	AN : LEGGINGS	M	Maroon
12	AN203-MAROON-S	AN203	2	AN : LEGGINGS	S	Maroon
13	AN203-MAROON-XL	AN203	2	AN : LEGGINGS	XL	Maroon
14	AN203-MAROON-XXL	AN203	3	AN : LEGGINGS	XXL	Maroon
15	AN204-PURPLE-L	AN204	10	AN : LEGGINGS	L	Purple
16	AN204-PURPLE-M	AN204	4	AN : LEGGINGS	M	Purple
17	AN204-PURPLE-S	AN204	6	AN : LEGGINGS	S	Purple
18	AN204-PURPLE-XL	AN204	3	AN : LEGGINGS	XL	Purple
19	AN204-PURPLE-XXL	AN204	5	AN : LEGGINGS	XXL	Purple



index	DATE	Months	CUSTOMER	Style	SKU	Size	PCS	RATE	GROSS AMT
0	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5004	MEN5004-KR-L	L	1	616.56	617
1	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5004	MEN5004-KR-XL	XL	1	616.56	617
2	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5004	MEN5004-KR-XXL	XXL	1	616.56	617
3	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5009	MEN5009-KR-L	L	1	616.56	617
4	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5011	MEN5011-KR-L	L	1	616.56	617
5	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5025	MEN5025-KR-L	L	1	649.03	649
6	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5015	MEN5015-KR-XL	XL	1	616.56	617
7	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5022	MEN5022-KR-XXL	XXL	1	649.03	649
8	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5014	MEN5014-KR-S	S	1	649.03	649
9	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5013	MEN5013-KR-S	S	1	649.03	649
10	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5013	MEN5013-KR-XXL	XXL	1	649.03	649
11	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5003	MEN5003-KR-L	L	1	616.56	617
12	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5003	MEN5003-KR-XXL	XXL	1	616.56	617
13	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5028	MEN5028-KR-L	L	1	649.03	649
14	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5028	MEN5028-KR-XXL	XXL	1	649.03	649
15	6/5/2021	6/21/2026	REVATHY LOGANATHAN	MEN5006	MEN5006-KR-XL	XL	1	649.03	649
16	6/5/2021	6/21/2026	REVATHY LOGANATHAN	BL097	BL097-M	M	1	324.35	324
17	6/5/2021	6/21/2026	REVATHY LOGANATHAN	BL104	BL104-M	M	1	421.85	422
18	6/5/2021	6/21/2026	REVATHY LOGANATHAN	BL104	BL104-XL	XL	1	421.85	422
19	6/5/2021	6/21/2026	REVATHY LOGANATHAN	BL103	BL103-S	S	1	389.35	389

Handling Missing & Null Values





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File | Home | Transform | Add Column | View | Tools | Help

Queries [4]
- Amazon Sale Report
- P L March 2021
- Sale Report
- International sale Report

```
= Table.ReplaceValue(#"Replaced Value1","","Unshipped",Replacer.ReplaceValue,{"Courier Status"})
```

	ship-country	promotion-ids	B2B	fulfilled-by	Unnamed: 22
Valid	100%	67%	100%	21%	0%
Error	0%	0%	0%	0%	0%
Empty	0%	33%	0%	79%	100%
	1 distinct, 0 unique	67 distinct, 30 unique	2 distinct, 0 unique	2 distinct, 0 unique	1 distinct, 0 unique

1	0081	IN		FALSE	Easy Ship
2	0085	IN	Amazon PLCC Free-Financing Universal Merchant AAT-WNKTBO3K27E...	FALSE	Easy Ship
3	0210	IN	IN Core Free Shipping 2015/04/08 23-48-5-108	TRUE	
4	5008	IN		FALSE	Easy Ship
5	0073	IN		FALSE	
6	1102	IN	IN Core Free Shipping 2015/04/08 23-48-5-108	FALSE	
7	0036	IN	IN Core Free Shipping 2015/04/08 23-48-5-108	FALSE	
8	0032	IN	Amazon PLCC Free-Financing Universal Merchant AAT-WNKTBO3K27E...	FALSE	Easy Ship
9	0008	IN	IN Core Free Shipping 2015/04/08 23-48-5-108	FALSE	
10	0041	IN		FALSE	
11	0073	IN		FALSE	
12	1303	IN		FALSE	
13	4606	IN	Amazon PLCC Free-Financing Universal Merchant AAT-WNKTBO3K27E...	FALSE	Easy Ship
14	0053	IN		FALSE	
15	0053	IN	Amazon PLCC Free-Financing Universal Merchant AAT-WNKTBO3K27E...	FALSE	Easy Ship
16	5801	IN	Amazon PLCC Free-Financing Universal Merchant AAT-WNKTBO3K27E...	FALSE	Easy Ship
17	2020	IN	IN Core Free Shipping 2015/04/08 23-48-5-108	FALSE	
18	0074	IN		FALSE	

Column statistics | Value distribution

Empty distribution.

COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Queries [4]
- Fact_Sales
- Dim_Product_Pricing
- Dim_Stock
- Dim_International

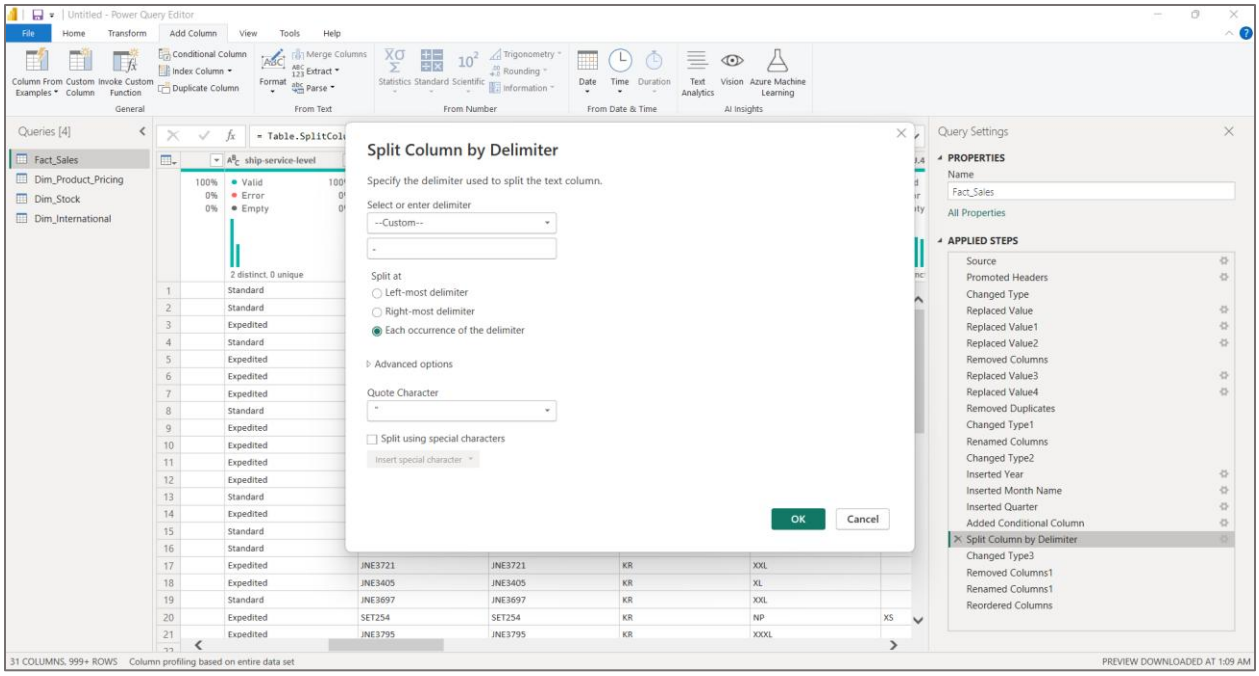
```
= Table.Distinct(#"Replaced Value4")
```

Date: 1.2 Decimal Number; $ Fixed decimal number; 123 Whole Number; % Percentage; Date/Time; Date; Time; Date/Time/Timezone; Duration; Text; True/False; Binary; Using Locale...

	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level
Valid	100%		100%	100%	100%	100%
Error	0%		0%	0%	0%	0%
Empty	0%		0%	0%	0%	0%
	128975 uniq...		distinct, 1 unique	2 distinct, 0 unique	2 distinct, 0 unique	2 distinct, 0 unique

1	0	405-8078784-5731545	celled	Merchant	Amazon.in	Standard
2	1	171-9198151-1101146	ped - Delivered to Buyer	Merchant	Amazon.in	Standard
3	2	404-0687676-7273146	ped	Amazon	Amazon.in	Expedited
4	3	403-9615377-8133951	celled	Merchant	Amazon.in	Standard
5	4	407-1069790-7240320	ped	Amazon	Amazon.in	Expedited
6	5	404-1490984-4578765	ped	Amazon	Amazon.in	Expedited
7	6	408-5748499-6859555	ped	Amazon	Amazon.in	Expedited
8	7	406-7807733-3785945	Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard
9	8	407-5443024-5233168	4/30/2022 Cancelled	Amazon	Amazon.in	Expedited
10	9	402-4393761-0311520	4/30/2022 Shipped	Amazon	Amazon.in	Expedited
11	10	407-5633625-6970741	4/30/2022 Shipped	Amazon	Amazon.in	Expedited
12	11	171-4638481-6326716	4/30/2022 Shipped	Amazon	Amazon.in	Expedited
13	12	405-5513694-8146768	4/30/2022 Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard
14	13	408-7955685-3083534	4/30/2022 Shipped	Amazon	Amazon.in	Expedited
15	14	408-1298370-1920302	4/30/2022 Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard
16	15	403-4965581-9520319	4/30/2022 Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard
17	16	406-9379318-6555504	4/30/2022 Shipped	Amazon	Amazon.in	Expedited
18	17	405-9013803-8009918	4/30/2022 Shipped	Amazon	Amazon.in	Expedited
19	18	402-4030358-5835511	4/30/2022 Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard
20	19	405-5957858-1051546	4/30/2022 Shipped	Amazon	Amazon.in	Expedited
21	20	405-0607769-0716360	4/30/2022 Shipped	Amazon	Amazon.in	Expedited

23 COLUMNS, 999+ ROWS Column profiling based on entire data set

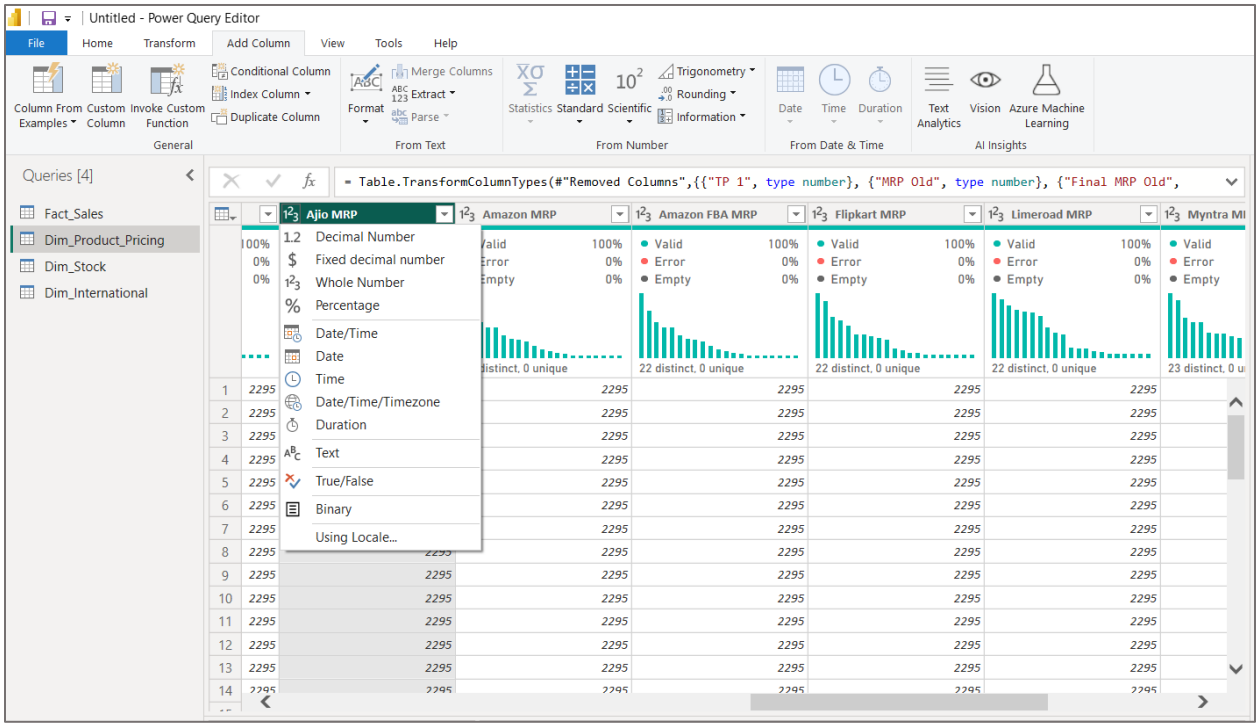


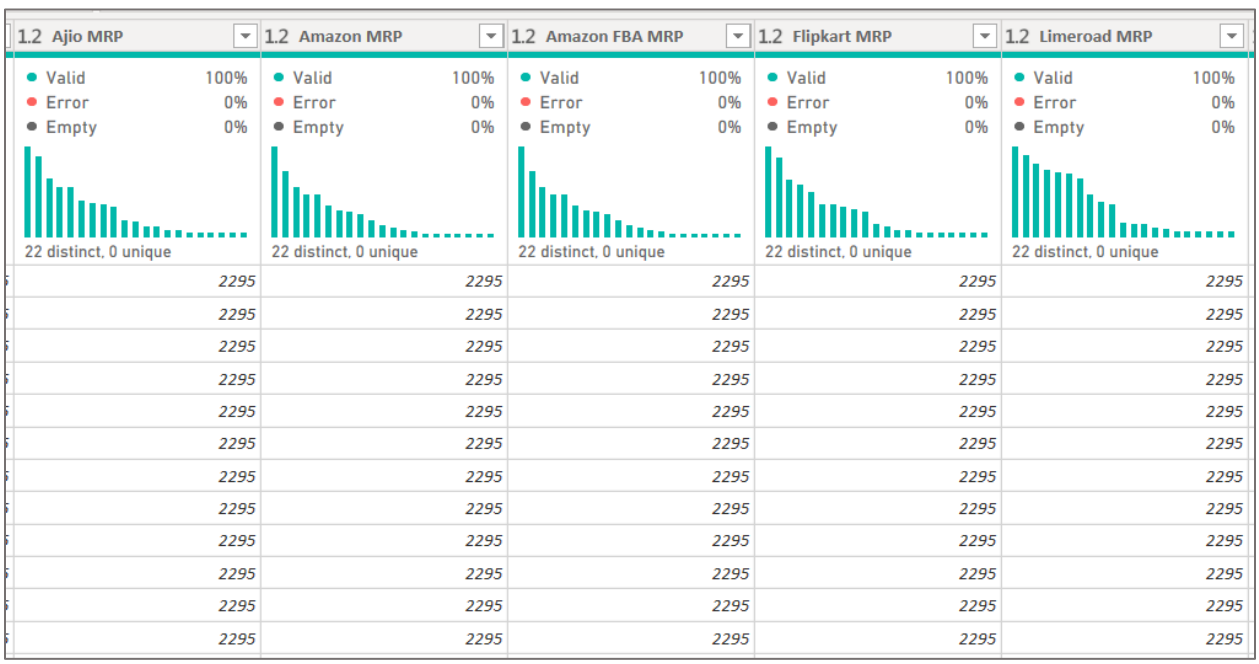
Renaming Columns

Product_Style_Code	Product_Category_Code	Size_Attribute
Valid 100%	Valid 99%	Valid 99%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty < 1%	Empty < 1%
1377 distinct, 117 unique	76 distinct, 11 unique	74 distinct, 9 unique
SET389	KR	NP
JNE3781	KR	XXXL
JNE3371	KR	XL
J0341	DR	L
JNE3671	TU	XXXL
SET264	KR	NP
J0095	SET	L
JNE3405	KR	S
SET200	KR	NP
JNE3461	KR	XXL

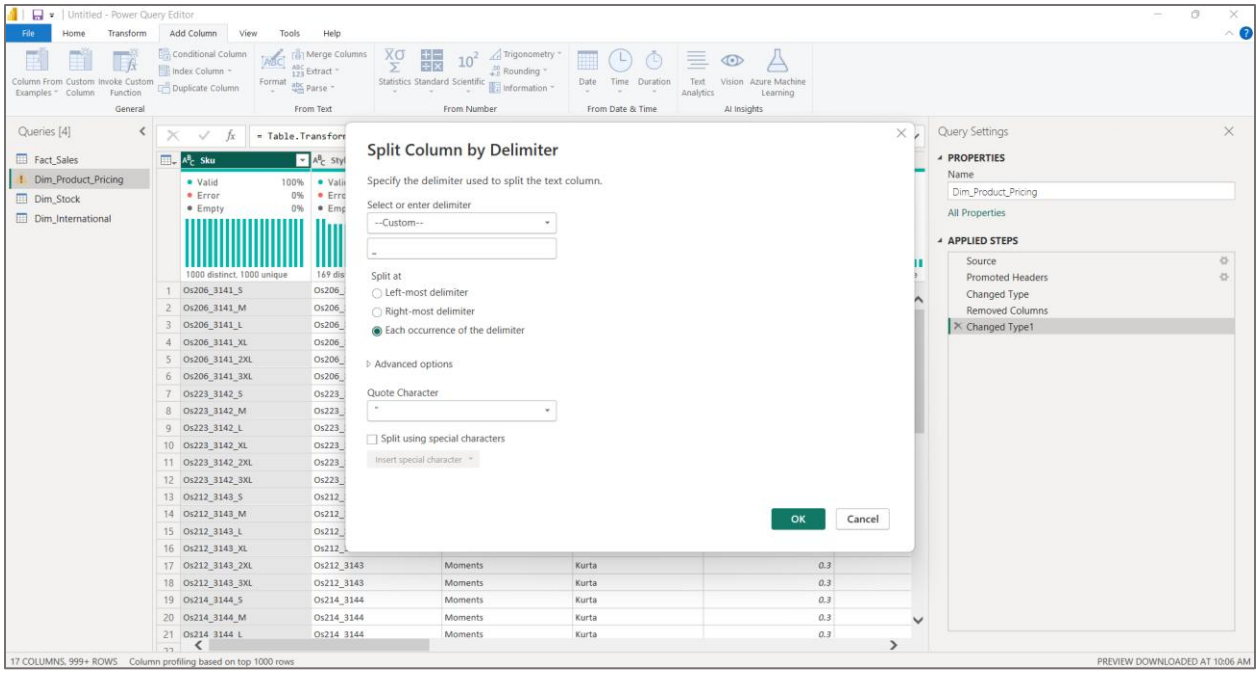
Second table transformation

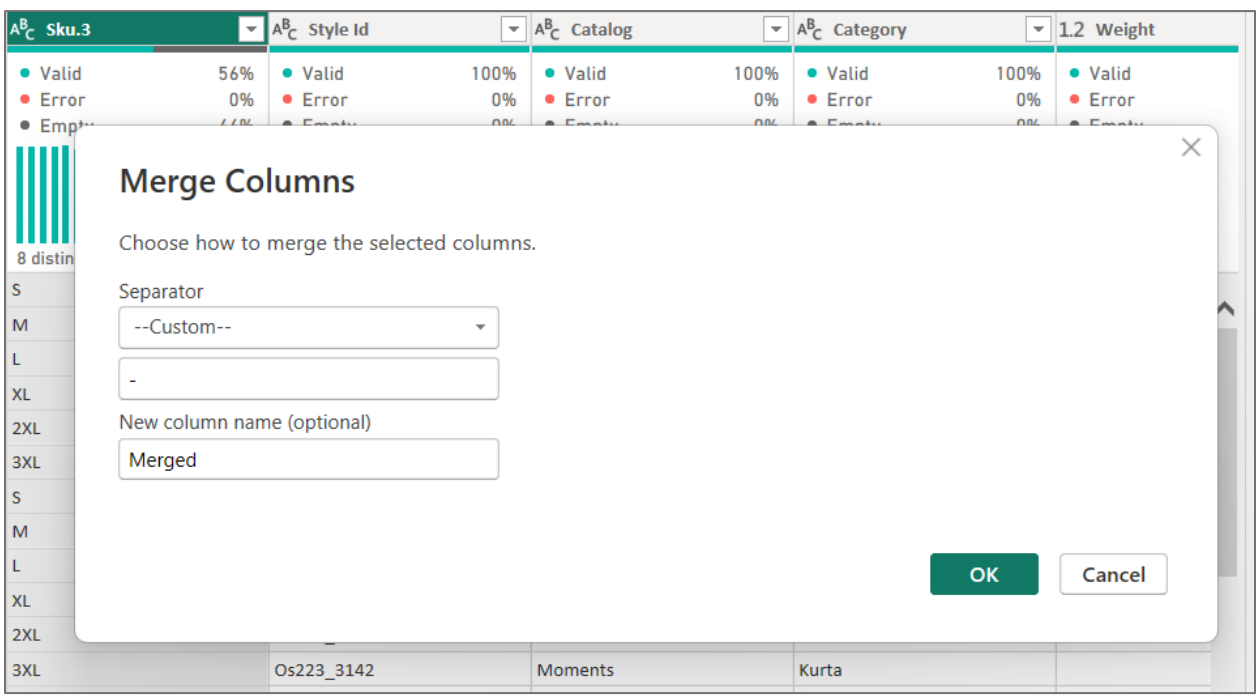
Changing Weight, TP 1, TP 2, and all the MRP columns to Fixed Decimal Number



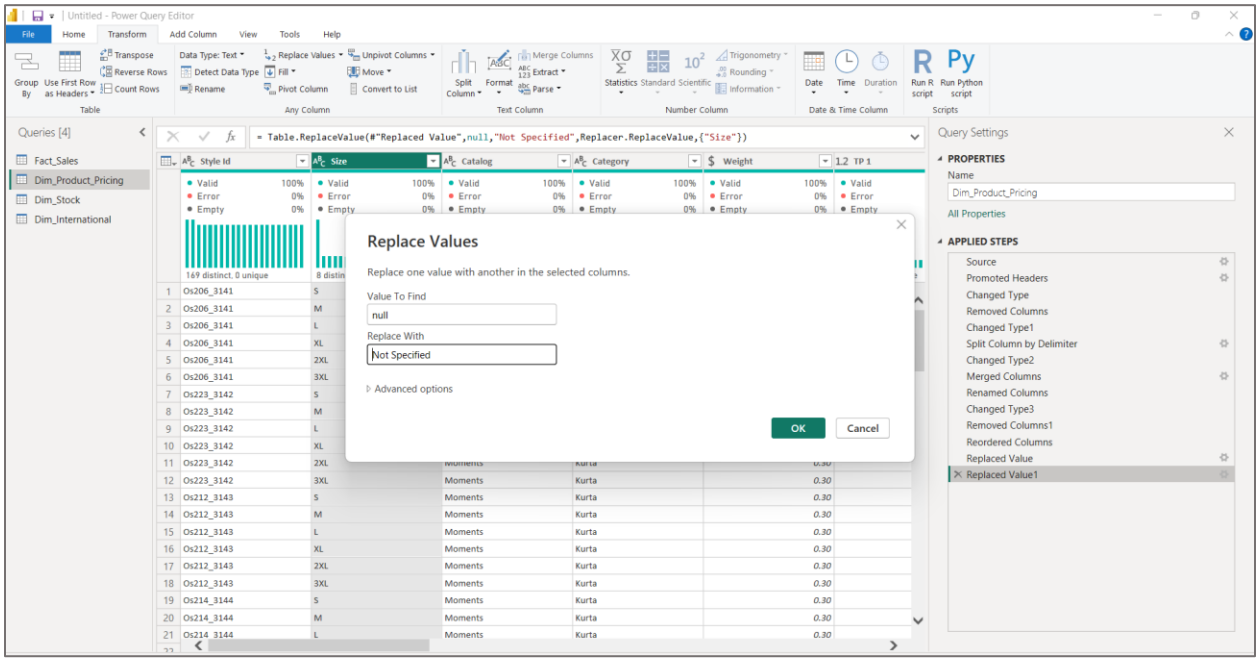


Splitting a column:


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [4]
Fact_Sales
Dim_Product_Pricing
Dim_Stock
Dim_International
Split Column by Delimiter
Specify the delimiter used to split the text column.
Select or enter delimiter
--Custom--
-
Split at
Left-most delimiter
Right-most delimiter
Each occurrence of the delimiter
Advanced options
Quote Character
"
Split using special characters
Insert special character
OK
Cancel
Query Settings
PROPERTIES
Name
Dim_Product_Pricing
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Removed Columns
Changed Type1
17 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 10:06 AM


Sku.3
Style Id
Catalog
Category
Weight
Valid
Error
Merge Columns
Choose how to merge the selected columns.
Separator
--Custom--
-
New column name (optional)
Merged
OK
Cancel
Os223_3142
Moments
Kurta

Handling missingness of table 2:



Query Settings

PROPERTIES

Name

Dim_Product_Pricing

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Columns
- Changed Type1
- Split Column by Delimiter
- Changed Type2
- Merged Columns
- Renamed Columns
- Changed Type3
- Removed Columns1
- Reordered Columns
- Replaced Value
- Replaced Value1

Query Settings

PROPERTIES

Name

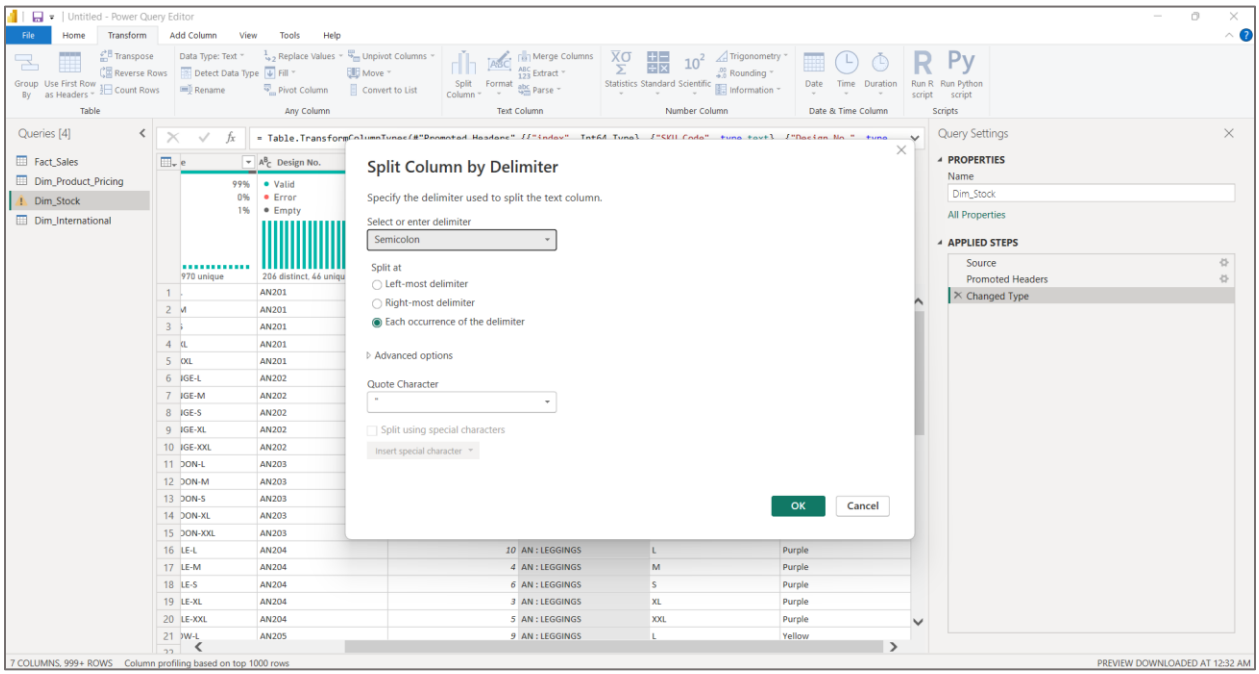
Fact_Sales

All Properties

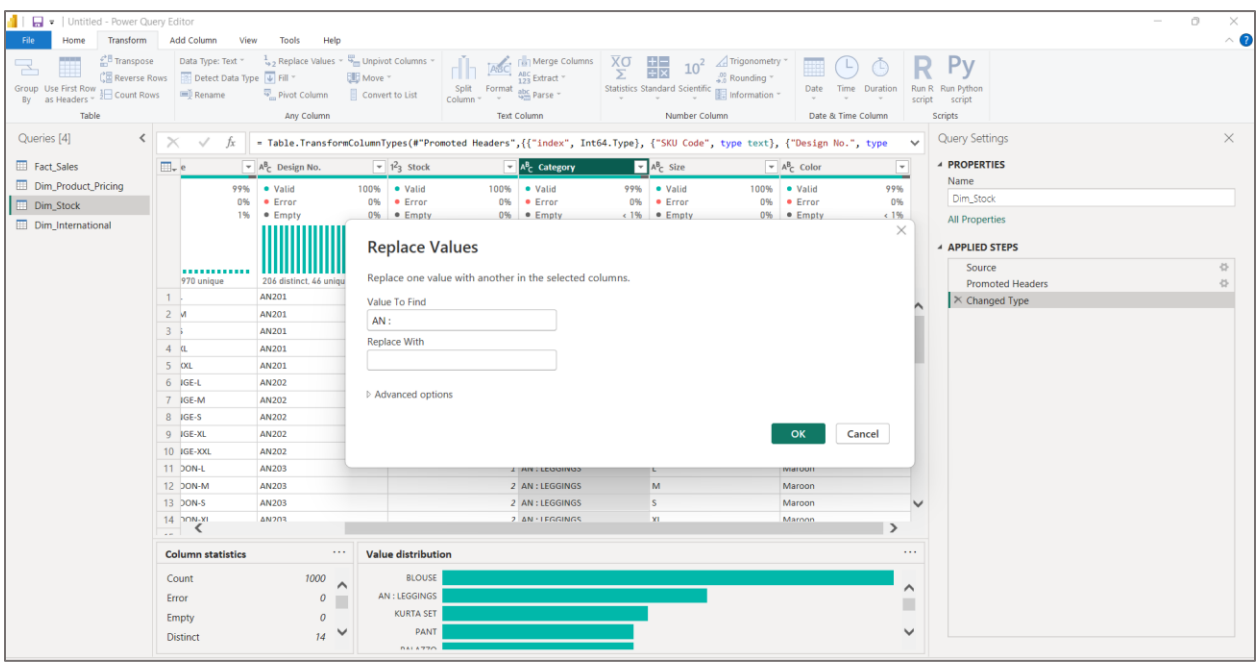
APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Replaced Value
- Replaced Value1
- Replaced Value2
- Removed Columns
- Replaced Value3
- Replaced Value4
- Removed Duplicates
- Changed Type1
- Renamed Columns
- Changed Type2
- Inserted Year
- Inserted Month Name
- Inserted Quarter
- Added Conditional Column
- Split Column by Delimiter
- Changed Type3
- Removed Columns1
- Renamed Columns1
- Reordered Columns

Transforming the 3rd table



Handling inconsistent entries



Displaying Applied steps of table 3

Query Settings

PROPERTIES

Name

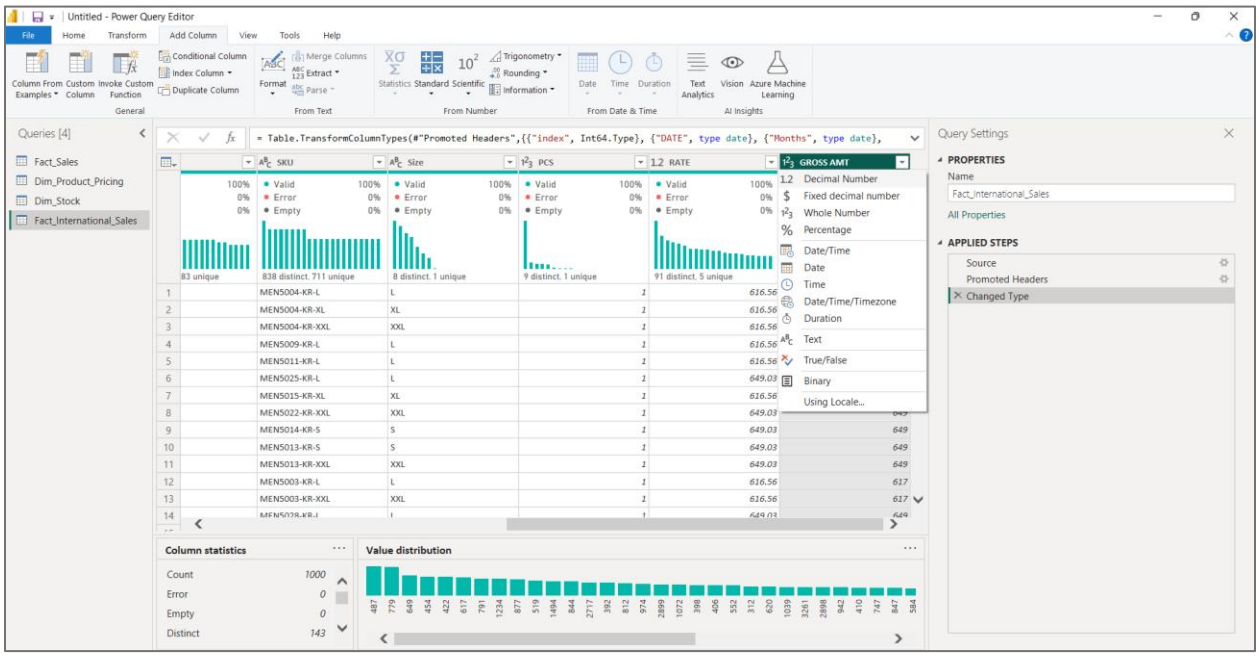
Dim_Stock

All Properties

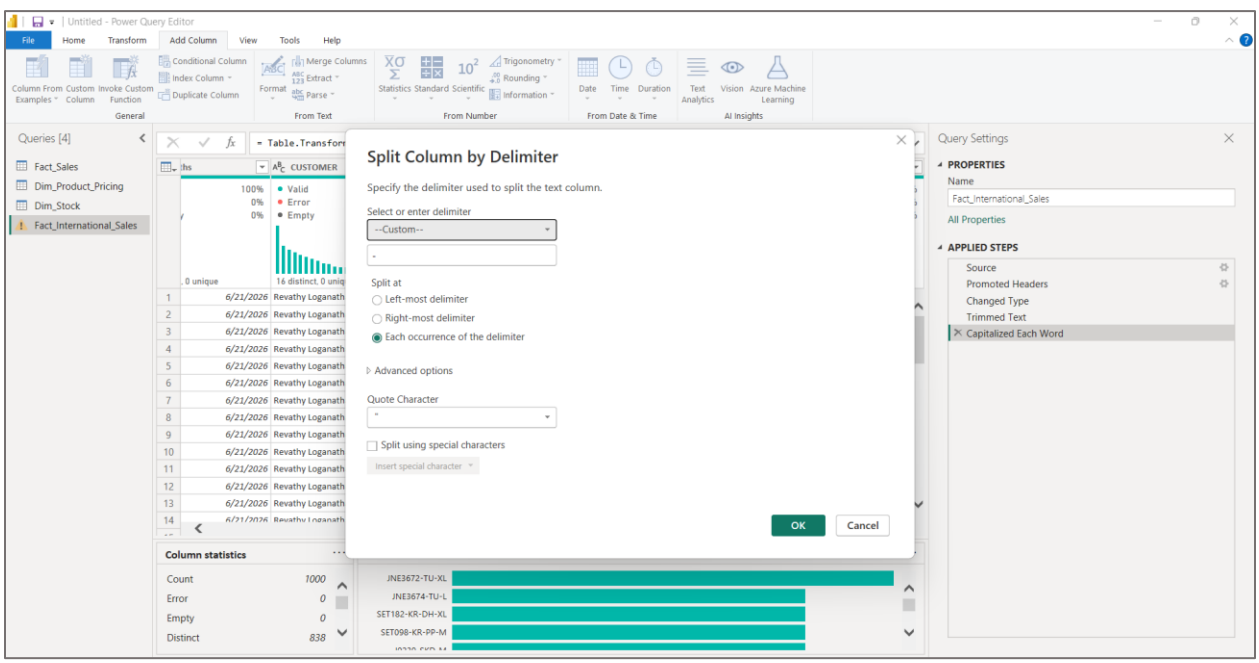
APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Replaced Value
- Capitalized Each Word
- Split Column by Delimiter
- Changed Type1
- Removed Columns
- Renamed Columns
- Trimmed Text
- Renamed Columns1
- Added Conditional Column
- Removed Columns1

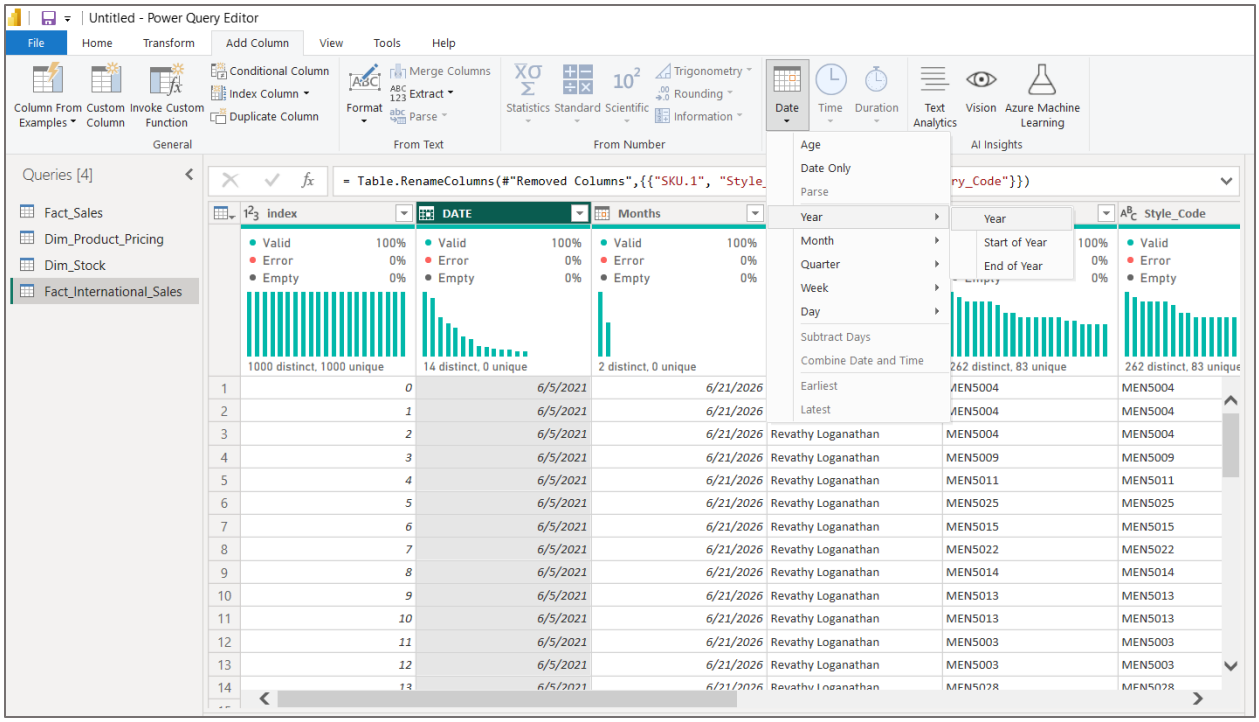
Transforming table 4



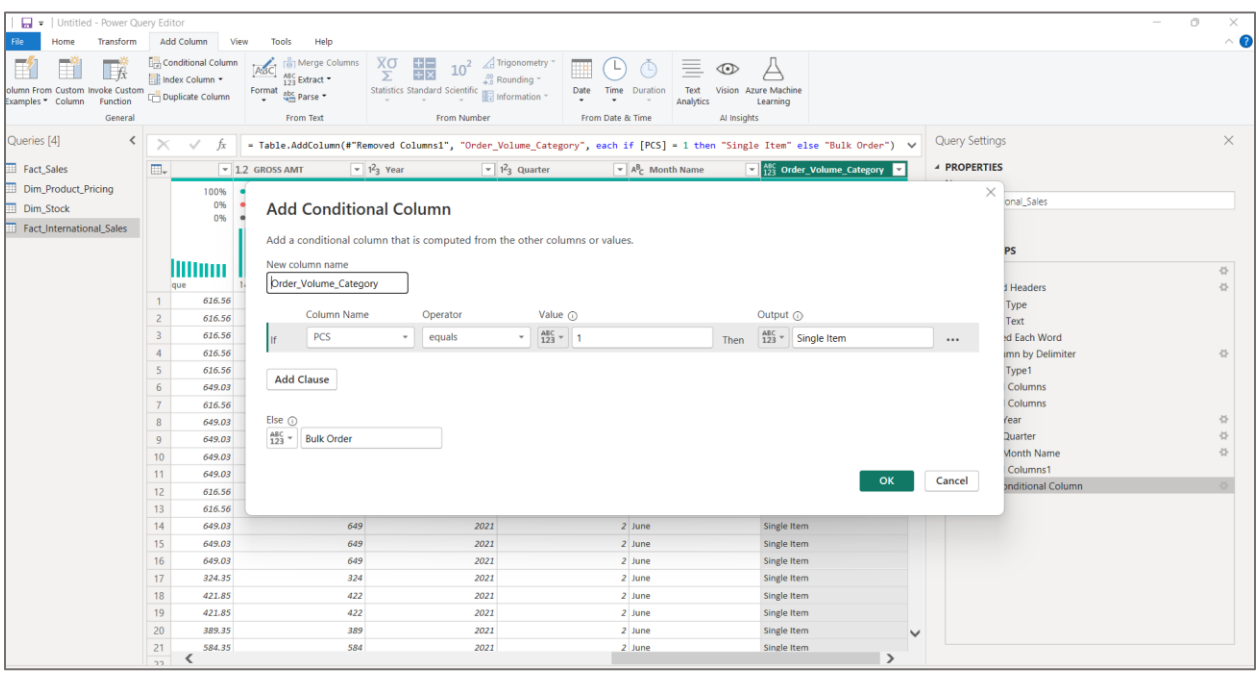
Splitting column



Time extraction



Creating a derived column



Applied steps of fourth table

Query Settings

PROPERTIES

Name

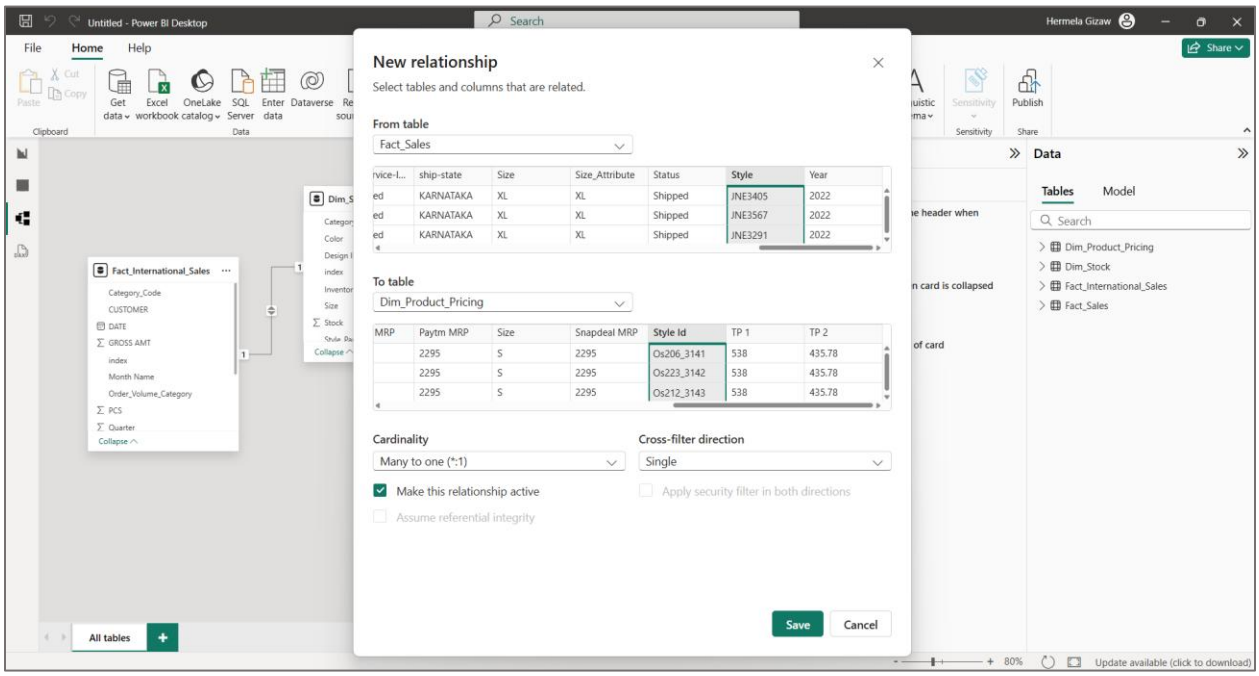
Fact_International_Sales

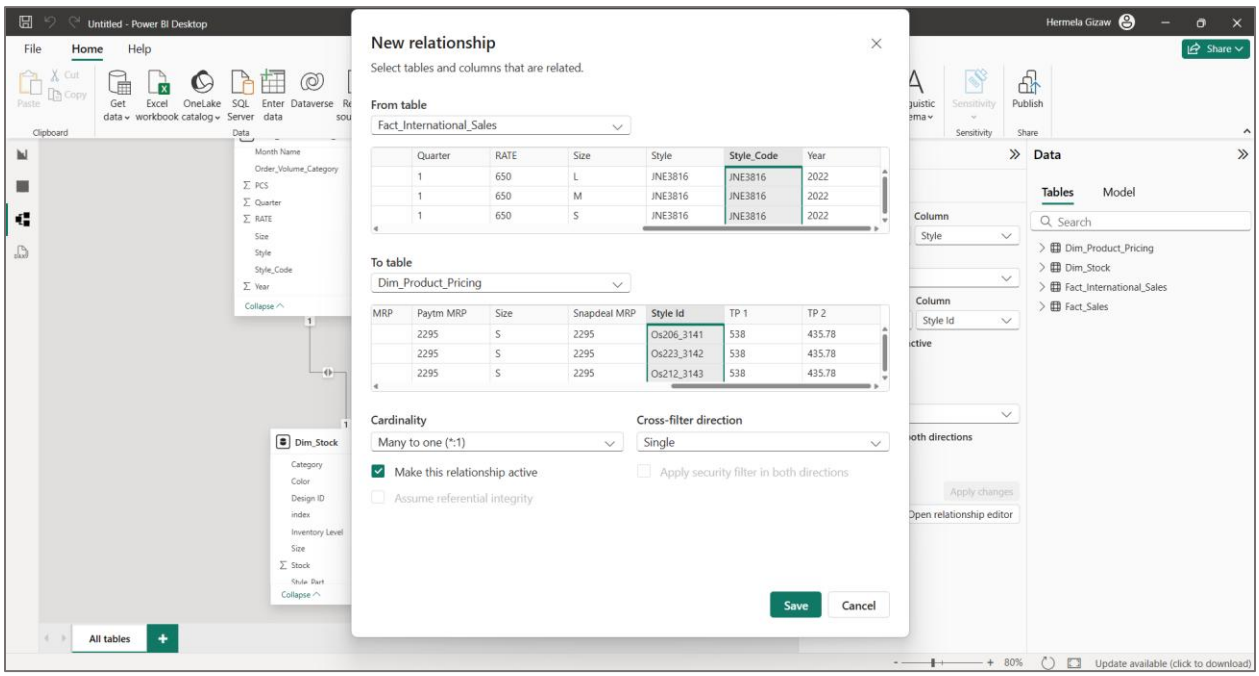
All Properties

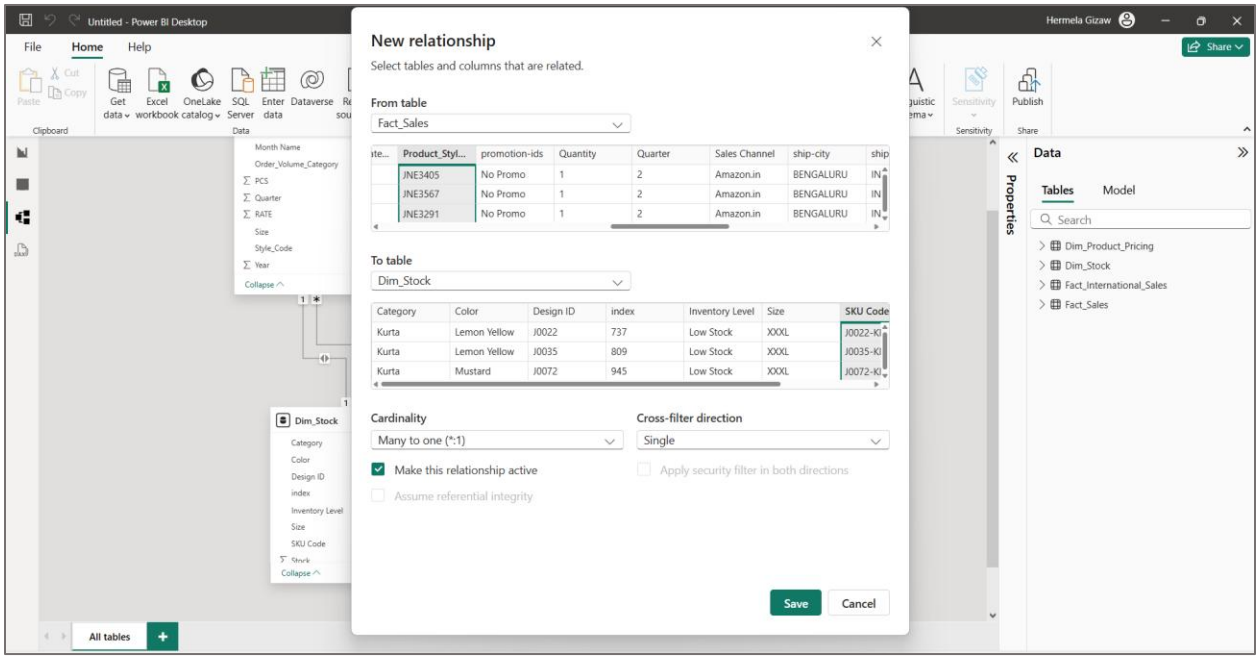
APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Trimmed Text
- Capitalized Each Word
- Split Column by Delimiter
- Changed Type1
- Removed Columns
- Renamed Columns
- Inserted Year
- Inserted Quarter
- Inserted Month Name
- Removed Columns1
- Added Conditional Column

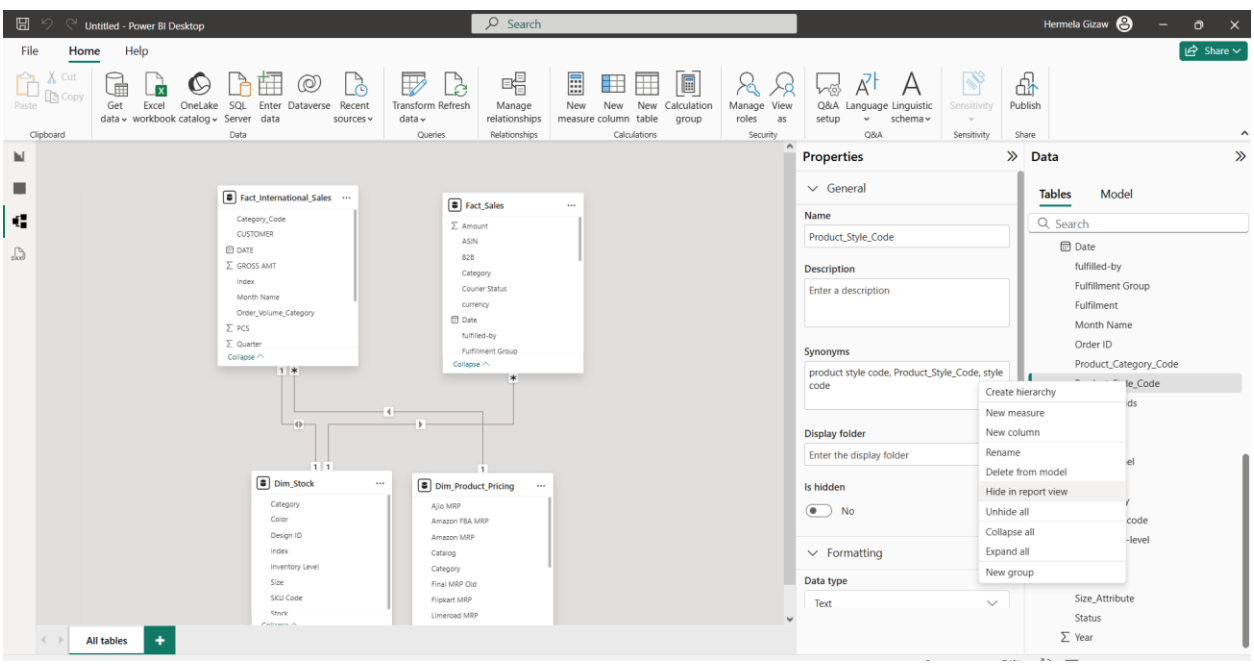
Data Modelling

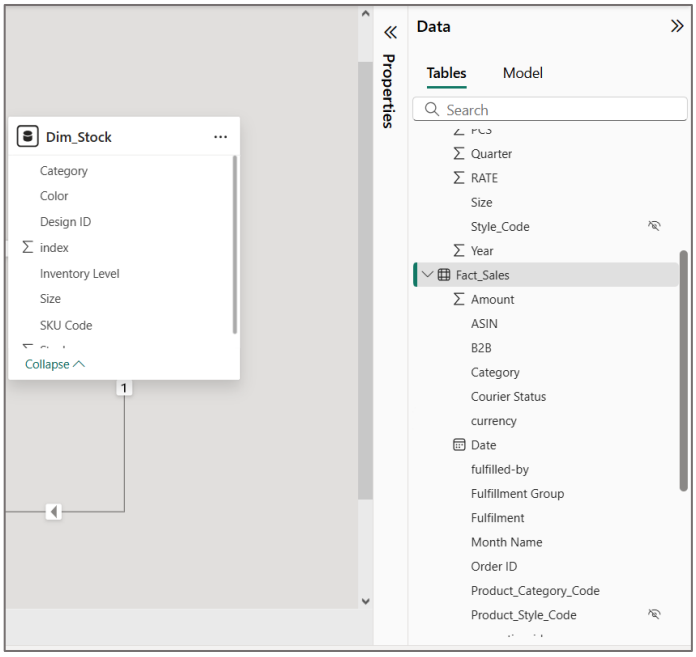




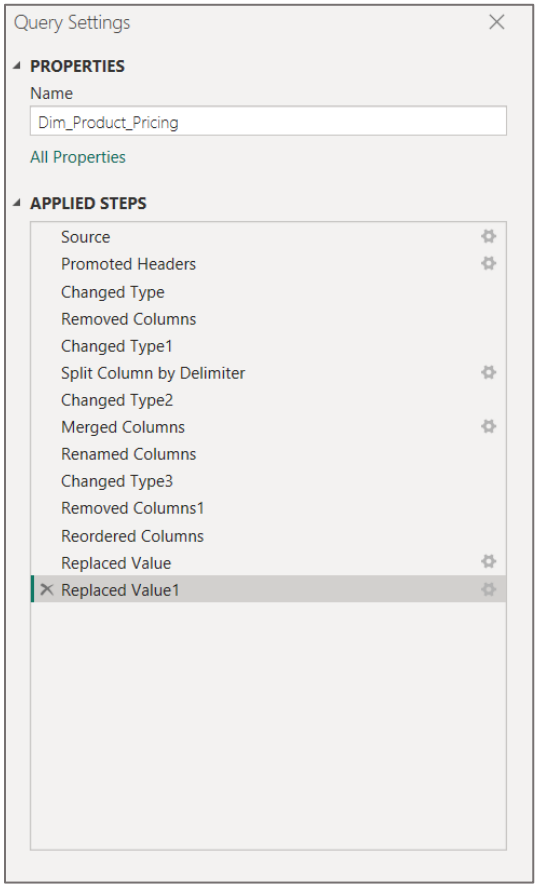


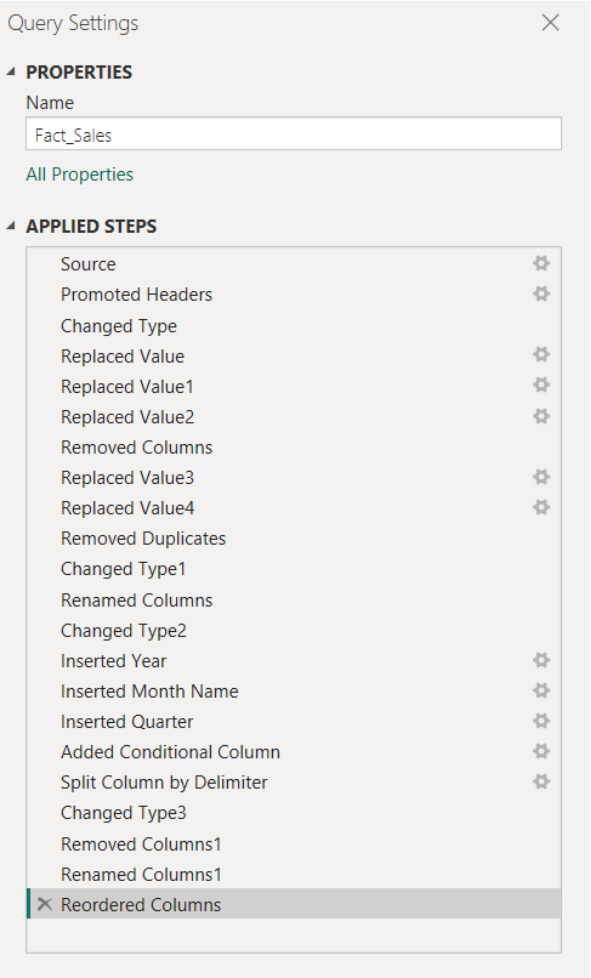
Hide in report view





Applied steps





Query Settings

PROPERTIES

Name

Dim_Stock

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Replaced Value
- Capitalized Each Word
- Split Column by Delimiter
- Changed Type1
- Removed Columns
- Renamed Columns
- Trimmed Text
- Renamed Columns1
- Added Conditional Column
- Removed Columns1

Query Settings

PROPERTIES

Name

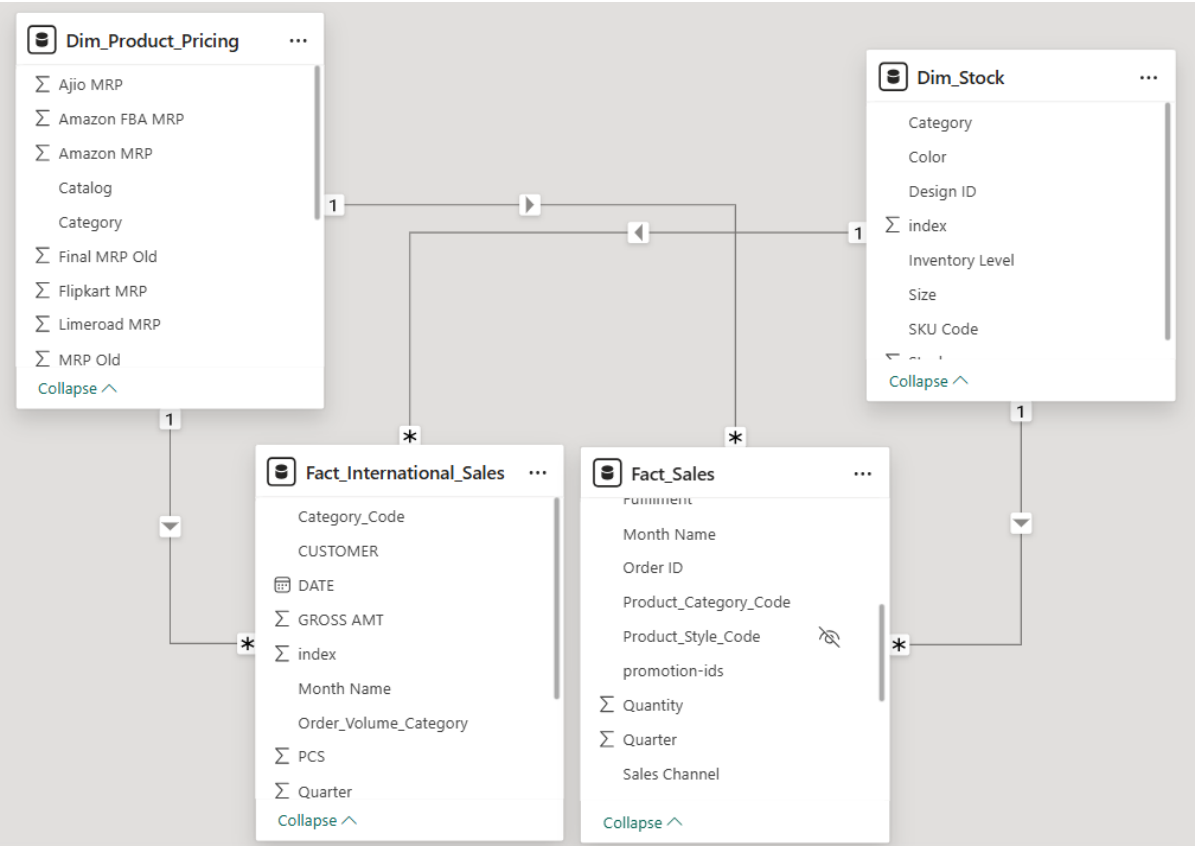
Fact_International_Sales

All Properties

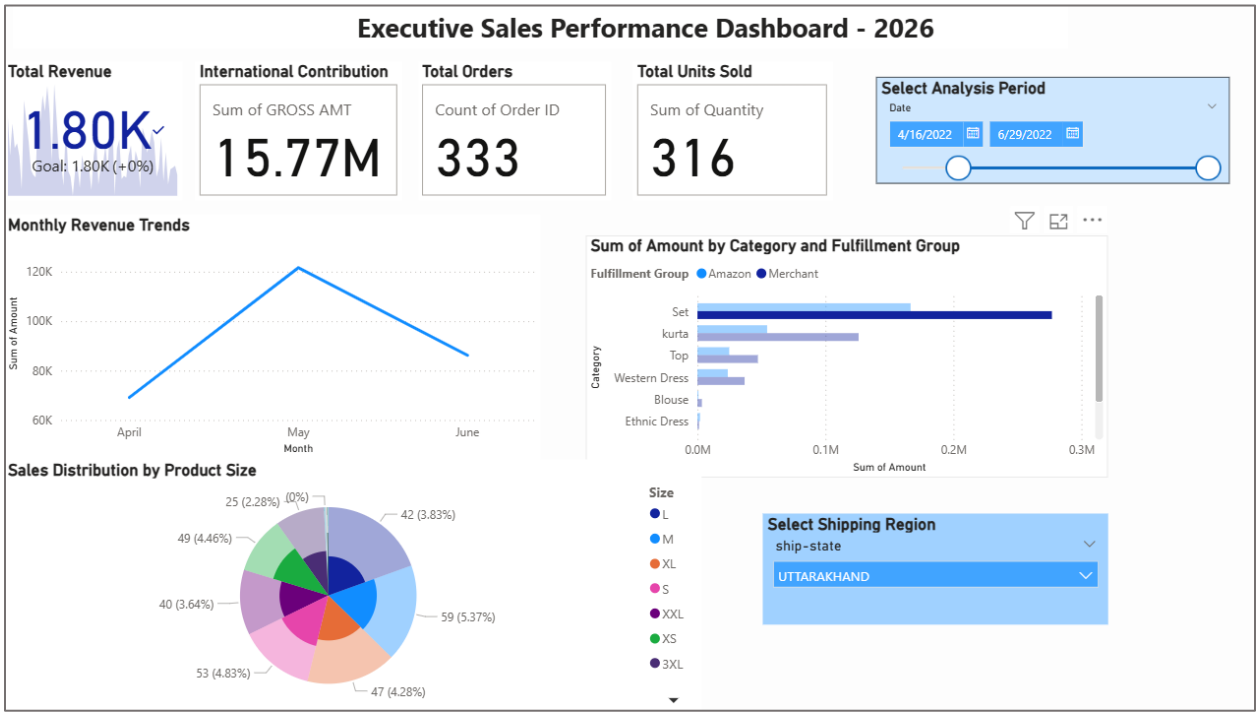
APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Trimmed Text
- Capitalized Each Word
- Split Column by Delimiter
- Changed Type1
- Removed Columns
- Renamed Columns
- Inserted Year
- Inserted Quarter
- Inserted Month Name
- Removed Columns1
- Added Conditional Column

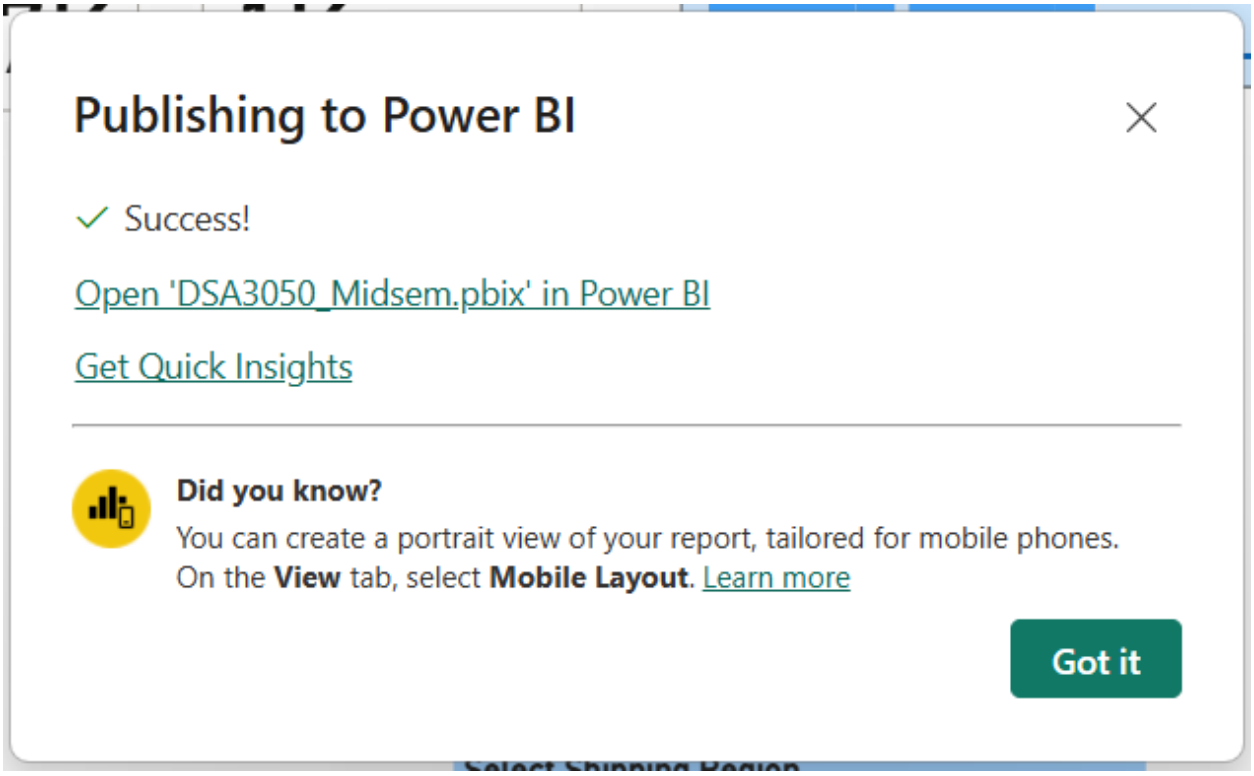
Data model relationships



Dashboard pages



Publishing confirmation



Section 2 - Dashboard Link

https://app.powerbi.com/view?r=eyJrIjoiYjU0ZWRlNWMtMWQ5My00M2YxLTk4N2EtNjRlODhlM2JjYmMyIiwidCI6IjE2ZDgzZWU2LTI1NGEtNDY5ZC1hNmNjLTU0ZTJjYTIzMTNlNyIsImMiOjh9

Section 3 - Github Link

HermelaDev/E-Commerce-Sales-Project-DSA3050